

An exact census of symmetric chain decompositions of $M(6)$

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Abstract

The poset of strict partitions with largest part at most six, ordered by ordinary Young-diagram containment, has exactly 1344 symmetric chain decompositions. This certifies equality in the previously reported lower bound. Rank sizes force five chains, with distinct starting ranks 0, 3, 5, 6, 9. Exhaustive exact cover over all 75 584 eligible symmetric paths produces a complete census and a recurrence certificate with 31 621 nonterminal states. A separate checker regenerates every branch, and an independently implemented rank-transfer enumeration reproduces the entire census. The result concerns this fixed poset, with no identification under automorphisms or duality.

1 Statement and context

Let $M(n)$ consist of the empty partition and all strict partitions $\lambda = (\lambda_1 > \cdots > \lambda_k > 0)$ with $\lambda_1 \leq n$. Its order is ordinary Young-diagram containment: $\lambda \leq \mu$ means $\lambda_i \leq \mu_i$ for every i , after padding with zeros. Its rank is $|\lambda| = \sum_i \lambda_i$ and its maximum rank is $N = n(n+1)/2$. A symmetric chain is a saturated chain whose endpoint ranks sum to N . A symmetric chain decomposition (SCD) is an unordered partition of all vertices into such chains.

Theorem 1. *On the fixed poset $M(6)$, the number of symmetric chain decompositions is*

$$\# \text{SCD}(M(6)) = 1344.$$

Here all 64 vertices, including the empty partition, are covered, and decompositions related by an automorphism or order-reversing duality are counted separately whenever they are distinct as partitions into chains.

Dorward [1, Figure 4] reports the lower bound 1344 for $M(6)$; Appendix A gives all decompositions for $M(3)$, $M(4)$, and $M(5)$. The accompanying program [2] already uses consecutive-rank matching and canonical storage, with stopping limits. The contribution here is an explicit upper-bound certificate and a durable complete census for $M(6)$. The number 1344 and the underlying enumeration strategies are not claimed as original. In particular, checking 1344 distinct examples alone would not establish Theorem 1. The comparison is limited to these inspected sources; no broad priority claim is made. As adjacent computational context, certified finite counting bounds for posets and exhaustive cover-based upper bounds occur in [3, 4]; these references are not used as results about symmetric chain decompositions.

2 Reduction to exact cover

Every vertex of $M(6)$ corresponds to a subset of $\{1, \dots, 6\}$, listed in decreasing order, so there are $2^6 = 64$ vertices. This subset description is an encoding of vertices, not the partial order. The rank generating polynomial is

$$\sum_{r=0}^{21} b_r q^r = \prod_{i=1}^6 (1 + q^i), \quad (1)$$

where

$$(b_0, \dots, b_{21}) = (1, 1, 1, 2, 2, 3, 4, 4, 4, 5, 5, 5, 5, 4, 4, 4, 3, 2, 2, 1, 1, 1).$$

The covers can be generated by increasing one part by one, preserving strictness and the upper bound six, or by appending a part one when permitted. To see completeness, suppose $\lambda < \mu$ and let i be their first unequal zero-padded coordinate. Increase λ_i by one. If $i > 1$ and this collided with the preceding part, then $\mu_{i-1} = \lambda_{i-1} = \lambda_i + 1$ and strictness of μ would imply $\mu_i \leq \lambda_i$, a contradiction. The increment stays below μ_i , or equals it, and the first zero coordinate simply becomes a part one. Thus an element ν exists with $\lambda < \nu \leq \mu$ and $|\nu| = |\lambda| + 1$. Consequently the covers are exactly the rank-one containment pairs. Direct construction gives 112 upward covers.

Lemma 1. *Every SCD of $M(6)$ has exactly one chain starting at each rank in $S = \{0, 3, 5, 6, 9\}$ and no other chains.*

Proof. Every symmetric chain starts at rank at most ten and ends at rank at least eleven. If a_s is the number starting at rank s , then $b_r = \sum_{s=0}^r a_s$ for $0 \leq r \leq 10$. Hence $a_s = b_s - b_{s-1}$, with $b_{-1} = 0$. Equation (1) gives precisely the stated starts. $\square$

For $s \in S$, let $\mathcal{C}_s$ be all directed cover paths from rank s to rank $21 - s$. Enumerating from every vertex at rank s and following every upward cover generates this set without omission. A path is determined by its vertex set, since rank fixes its order. The path counts are given in Table 1.

Starting rank s	0	3	5	6	9
Vertices per chain	22	16	12	10	4
$ \mathcal{C}_s $	33 592	33 592	6 398	1 958	44

Table 1: All eligible full symmetric paths in $M(6)$.

An SCD is exactly a selection of one path from each $\mathcal{C}_s$ with pairwise disjoint vertex sets. Indeed, necessity follows from Lemma 1. Conversely, disjoint paths of these lengths cover $22+16+12+10+4 = 64$ vertices. The distinct start ranks distinguish the chains intrinsically, so this selection introduces no factorial multiplicity into the count of unordered decompositions.

3 Exhaustive recurrence and its verification

For a residual vertex set U and a remaining set $R \subseteq S$, let $F(U, R)$ count disjoint choices of one path from each remaining group that cover exactly U . Put $F(\emptyset, \emptyset) = 1$ and $F(U, \emptyset) = 0$ when $U \neq \emptyset$. For any $g \in R$,

$$F(U, R) = \sum_{\substack{C \in \mathcal{C}_g \\ V(C) \subseteq U}} F(U \setminus V(C), R \setminus \{g\}). \quad (2)$$

For any $v \in U$, an alternative recurrence is

$$F(U, R) = \sum_{\substack{g \in R, C \in \mathcal{C}_g \\ v \in V(C) \subseteq U}} F(U \setminus V(C), R \setminus \{g\}). \quad (3)$$

Each formula partitions the solutions by a uniquely determined chain: the chain starting at g , or the chain containing v . Empty sums are zero. Each transition reduces $|R|$, which proves termination and justifies induction on the remaining groups.

The enumerator stores all 75 584 eligible paths. At each state it chooses a remaining group or uncovered vertex with the fewest eligible paths, breaking ties deterministically. After selecting a path it removes that group and every intersecting path. Inductively, the retained paths are exactly those in remaining groups contained in U . The pivot rule affects efficiency only; either recurrence is exact. There is no time cutoff, result cutoff, random sampling, or symmetry quotient in this computation.

For every distinct nonterminal state the certificate records U , R , the pivot, the number of branches, and the resulting integer count. The separate checker imports neither enumerator. It reconstructs the vertices and rank-one covers using zero-padded coordinate comparisons, then regenerates every eligible path at every reachable state. For a vertex pivot v , deleting the other vertices at its rank enforces passage through v : each saturated path meets every rank between its endpoints exactly once. A backward reachability calculation discards only vertices with no suffix to the required endpoint in that same residual graph.

The checker requires every nonterminal child record, checks the branch count and integer sum, validates residual rank occupancies and terminal coverage, and rejects duplicate or unreachable state records. Induction using (2)–(3) therefore proves the root value from the replay, including the upper bound. The state totals are reported in Table 2.

Number of selected chains	0	1	2	3	4
Distinct nonterminal states	1	44	2 406	12 820	16 350

Table 2: The 31 621 states of the fully replayed recurrence certificate.

The complete replay regenerates 38 203 branches. At the root the pivot is the start-nine group. Of its 44 paths, 36 have zero completions, two have 96 completions each, and six have 192 each. The checked root sum is therefore

$$F(M(6), S) = 36 \cdot 0 + 2 \cdot 96 + 6 \cdot 192 = 1344. \quad (4)$$

The saved census contains exactly 1344 distinct decompositions; each was also checked for coverage, saturation, and endpoint symmetry. Together, the exhaustive recurrence and its exact execution prove Theorem 1.

4 Independent transfer check and reproducibility

A further implementation generates strict partitions recursively and forms the order using explicit Young-diagram cell sets. It computes the transitive reduction and confirms the same 112 rank-one covers. At rank r , let P_r be the vertices and let $A_r = \{s \in S : s \leq r \leq 21 - s\}$ be the active chain starts. A state is a bijection $f : A_r \rightarrow P_r$. Two successive states f, g are compatible exactly when

$f(s) < g(s)$ for all shared labels. With weight one at the unique rank-zero state, compute

$$w_r(g) = \sum_{f \text{ compatible with } g} w_{r-1}(f).$$

All bijections are included, even those with zero weight. Every compatible history gives an SCD and conversely, because each label occupies exactly its prescribed symmetric rank interval. There are at most $5! = 120$ states per rank. The total history counts through ranks zero to 21 are

$$1, 1, 1, 2, 2, 6, 24, 24, 48, 288, 288, 576, 576, 672, \\ 1344, 1344, 1344, 1344, 1344, 1344, 1344, 1344.$$

Backtracking all histories produces the same complete set of 1344 decompositions as the exact-cover enumeration. This checks the census by a separately implemented exhaustive upper-bound calculation.

The accompanying archive `supplement.zip` contains the enumerator, certificate checker, transfer program, full recurrence certificate, canonical census, and a reproduction driver. After extraction, run

```
python3 reproduce.py
```

from the extracted directory, with Python 3.10 or later and assertions enabled (do not use `-O`). The programs use only the standard library; the tested interpreter is Python 3.13.5. The driver replays the certificate, reruns both counting methods, and compares all canonical witnesses. It writes only to a separate output directory.

A partition is serialized as the integer $m(\lambda) = \sum_{a \in \lambda} 2^{a-1}$. In the census each chain is listed in increasing rank, chains are sorted lexicographically, and the decompositions are likewise sorted. The recurrence instead encodes residual sets as bit masks relative to its explicit vertex list; its remaining-group indices refer to the ordered list $(0, 3, 5, 6, 9)$. These two encodings are distinguished in the archive documentation.

Scope and computational assistance. The result certifies one finite equality. It gives no formula for general n and does not resolve the general SCD existence question in [1, Question 1.1]. Its computational dependencies are the stated recurrences and correct execution of the supplied exact integer Python programs; no proof-assistant formalization was performed. OpenAI Codex assisted in developing the programs, checking the mathematical arguments, and preparing the manuscript. A separate model context reviewed the argument, replayed the certificate, and wrote the independent transfer implementation. This separation is a verification procedure, not external peer review or statistical independence.

References

- [1] Robert Dorward, *On the number of symmetric chain decompositions of the minuscule lattices $L(m, n)$ and $M(n)$* , arXiv:2606.15891v1 (2026).
- [2] Companion code to [1], `Mn_small_examples.py`, *symmetric-chain-decomposition* repository, inspected 8 September 2026. <https://github.com/google-research/symmetric-chain-decomposition>.
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- [4] Zijian Zeng, *The Strict-pair Product Extremal Value at 62 is 31* (2026). doi:10.2139/ssrn.7380458.